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Integrated Detector Support for Mobile X-Ray Device

Abstract

A mobile x-ray device includes a body including a number of wheels rotatably attached thereto and housing operative systems for the obtaining and processing of x-ray data to provide x-ray images, a moveable vertical column and horizontal arm operably connected to the body and including an x-ray emitter opposite the body, a detector configured to receive x-rays from the x-ray emitter and operably connected to the operative systems to transmit x-ray data from the detector to the operative systems, a drive handle disposed on the body, and a detector support having a sleeve defining an interior adapted to receive an edge of a detector therein, and a pair of securing plates extending outwardly from the sleeve and secured to the drive handle. The sleeve can securely engage opposed surfaces of the detector to allow the user/technician to readily position a protective bag around the exposed portion of the detector in a hands-free manner.

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Background/Summary

FIELD OF THE DISCLOSURE

[0001] The present disclosure is related to x-ray imaging devices, and more particularly to detector management supports for mobile x-ray imaging devices.

BACKGROUND OF THE DISCLOSURE

[0002] Digital imaging technologies including medical imaging technologies such as x-ray imaging allow for non-invasive acquisition of images of internal structures or features of a subject, such as a patient. Digital imaging systems produce digital data which can be reconstructed into radiographic images. In digital x-ray imaging systems, radiation from a source is directed toward the subject. A portion of the radiation passes through the subject and impacts a detector. The detector includes an array of discrete picture elements or detector pixels and generates output signals based upon the quantity or intensity of the radiation impacting each pixel region. The output signals are subsequently processed to generate an image that may be displayed for review. These images are used to identify and/or examine the internal structures and organs within a patient's body.

[0003] In order to facilitate obtaining x-ray images in a variety of locations and/or situation a number of mobile x-ray devices have been developed. These mobile x-ray devices have particular applicability in situations where the patient cannot or is not readily moveable, such that it is required to move the imaging device to the patient. The construction of the mobile x-ray device includes a mobile chassis that supports the entire x-ray assembly and a telescopic or fixed vertical column that is mounted on the chassis and can rotate about its long, vertical axis. The telescopic column includes a fixed portion jointed to the chassis, and one or more telescopic portions that are movably mounted to the fixed portion. The vertical column also includes a horizontal telescopic arm that can move along the column and that supports an x-ray emitter or source opposite the column.

[0004] In operation the mobile chassis can be moved into a location adjacent the patient or other object to be imaged. Then, employing the mechanisms disposed within the column and the telescopic arm, and the x-ray emitter can be moved to the desired position in order to obtain the x-ray images of the patient. The mechanisms allow the column and telescopic arm to be moved and counterbalanced by the remainder of the mobile x-ray device, thereby enabling the x-ray emitter to be properly and stably positioned to obtain the desired x-ray images.

[0005] The mobile x-ray device additionally includes one or more detectors associated with the mobile x-ray device. These detectors are capable of receiving the x-rays from the x-ray emitter that are directed through the patient and transmitting the information to the processing unit within the mobile chassis for generation of an x-ray image. The generated x-ray image can be presented on a display located on the mobile chassis in order to determine a proper course of treatment for the patient.

[0006] The detectors used in the imaging procedures performed with the mobile x-ray imaging device can be stored within a bin disposed on the chassis opposite the telescopic portion. The bin has dimensions that approximate the dimensions of the detectors, i.e., to enclose at least 50% to 67% of the area of the detector therein, such that the detectors can be securely retained within the bin during movement of the mobile x-ray device. The bin is disposed adjacent a bottom edge of the chassis to maintain the bin and detectors therein in an easy to access location that does not interfere with the operation and/or movement of the mobile x-ray device by the operator.

[0007] The detectors used with the mobile x-ray device can be operably connected by a wire or tether, or can be wirelessly connected to the processing device within the mobile chassis in order to transmit information and image data between the detector and the mobile chassis. To facilitate the handling of the detectors to place the detector in the desired position relative to the patient to obtain the desired image data to generate the x-ray image, the detector being used can be positioned within a grid handle or other removeable housing to provide a protective and more easily handled frame for the detector.

[0008] In use, as the detector is often placed in direct contact with the patient, to provide an amount of protection to the detector when in use, the detector and grid handle are placed within a protective bag or enclosure. The bag includes an open end through which the detector is inserted into the bag, and which is closed in a suitable manner when the detector is properly positioned within the bag.
[0009] The process for safely handling and inserting the detector or detector with grid/grid handle to place the detector within the bag prior to use can present several challenges. More specifically, due to the size of the detector, which can be up to 17″×17″ in size and weight up to 12 pounds, on many occasions the operator must balance the detector on a surface with one hand while placing the bag around the detector using the opposite hand. For example, the process of bagging the detector often involves placing the detector on the sloped top cover of the mobile x-ray device including the display, or on a near-by surface, such as a body part of the operator. In these scenarios, there is a significant potential for the detector to be placed in a position where it is not stable and to slide off of the surface leading to a detector drop event with the consequent negative effects leading to potential image quality issues or worse, detector failure. Further, the act of positioning and/or dropping the detector on the surface of the mobile x-ray device, i.e., on the display, can have negative consequences for the mobile x-ray device itself, e.g., damaging the display of the mobile x-ray device.
[0010] Additionally, the process of bagging the wireless or tethered detector is often a chaotic/acrobatic dance of holding the detector in one hand, bag in the other, balancing the detector on a body part, e.g., a knee for support of the detector and ultimately getting the bag onto the detector without dropping the detector. The operator may also rest the detector against their feet/shins to place the detector in the bag prior to use. The same scenarios outlined above often also occur when attempting to assemble a grid handle around a detector or remove the detector from it, or when removing the detector from within a bag after using in an imaging procedure. Each of these situations creates a poor ergonomic condition, and each situation can occur multiple times throughout the course of a single work shift, thereby creating the potential for injury to the operator.
[0011] Along with the issues created by the requirement for bagging the detector prior to each use in an imaging procedure, the mobile x-ray device includes an exposure switch that is connected to the chassis in order to enable the operator to selectively operate the x-ray emitter and obtain x-ray data from the detector in an imaging procedure. The exposure switch is connected to the chassis by an elongate cord that has a length sufficient to enable the operator to move a sufficient distance from the mobile x-ray device when the x-ray emitter is operated.
[0012] However, as a result of the length of the cord and/or the location of the connection of the cord to the chassis near the bin, the cord can often become positioned and tangled with the detectors in the bin requiring the operator to move the cord out of the way to position the detector and associated grid handle into or out of the bin. In addition, the exposure switch cord can also become entangled within a drive wheel of the mobile x-ray device or simply drag on the floor, neither of which is a desirable situation.
[0013] In order to alleviate these issues with the detectors, certain prior art mobile x-ray devices have incorporated notches within the chassis that accommodate the corners of the detector. These notches allow the corners of the detector to be inserted therein to function as the support for the detector on the chassis when a bag is positioned over or removed from the detector.
[0014] However, the location and sizes of the notches, either within a top surface of the chassis immediately adjacent the display or on the drive handle, only allow for the engagement with detectors of specific sizes, and only with or without grid handles attached. The notches can only securely engage the corners having a specified detector width, either with or without a grid handle, but not both. Also, the engagement of only the corners of the detector by the notches requires precise placement of both corners within the notches to effectively support the detector therein. In addition, the significantly limited areas of the detectors that can be engaged within the corners

enable the corners to be easily disengaged from within the corners, particularly when the detector is inadvertently contacted, which can readily result in dropping the detector.

[0015] As a result, it is desirable to develop a detector support for a mobile x-ray device that has a simplified construction that enables effective and reliable engagement of detectors of various sizes therein that avoids the shortcomings of the prior art.

SUMMARY OF THE DISCLOSURE

[0016] According to one aspect of an exemplary embodiment of the disclosure, a detector support for a mobile x-ray device includes a sleeve defining an interior adapted to receive an edge of a detector with or without a grid handle therein, and a pair of securing plates extending outwardly from the sleeve and adapted to be secured to the mobile x-ray device.

[0017] According to another exemplary embodiment of the disclosure, a method of supporting a detector on a mobile x-ray device includes the steps of providing a mobile x-ray device including a body including a number of wheels rotatably attached thereto and housing operative systems for the obtaining and processing of x-ray data to provide x-ray images, a moveable vertical column and horizontal arm operably connected to the body and including an x-ray emitter opposite the body, a detector configured to receive x-rays from the x-ray emitter and operably connected to the operative systems to transmit x-ray data from the detector to the operative systems, a drive handle disposed on the body, and a detector support having a sleeve defining an interior adapted to receive an edge of a detector therein, and a pair of securing plates extending outwardly from the sleeve and secured to the drive handle, placing an edge of the detector with or without a grid handle within the sleeve to support the detector within the sleeve in a hands-free manner, and placing a protective bag around the detector.

[0018] According to still another aspect of an exemplary embodiment of the present disclosure, a mobile x-ray device includes a body including a number of wheels rotatably attached thereto and housing operative systems for the obtaining and processing of x-ray data to provide x-ray images, a moveable vertical column and horizontal arm operably connected to the body and including an x-ray emitter opposite the body, a detector configured to receive x-rays from the x-ray emitter and operably connected to the operative systems to transmit x-ray data from the detector to the operative systems, a drive handle disposed on the body, and a detector support having a sleeve defining an interior adapted to receive an edge of a detector with or without a grid handle therein, and a pair of securing plates extending outwardly from the sleeve and secured to the drive handle.

[0019] These and other exemplary aspects, features and advantages of the invention will be made apparent from the following detailed description taken together with the drawing figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] The drawings illustrate the best mode currently contemplated of practicing the present invention.

[0021] In the drawings:

[0022] FIG. **1** is a side elevation view of a mobile x-ray device, according to an exemplary embodiment of the disclosure.

[0023] FIG. **2** is a rear elevation view of the mobile x-ray device of FIG. **1** with a detector support disposed on the drive handle for the device, according to an exemplary embodiment of the disclosure.

[0024] FIG. **3** is a partially broken away, isometric view of the drive handle for the mobile x-ray device of FIG. **1** with the detector support attached thereto, according to an exemplary embodiment of the disclosure.

[0025] FIG. **4** is an isometric view of the detector support of FIG. **3**, according to an exemplary

embodiment of the disclosure.
[0026] FIG. **5** is a cross-sectional view along line **5-5** of FIG. **4**.
[0027] FIG. **6** is an isometric view of the detector support of FIG. **3**, according to another exemplary embodiment of the disclosure.
[0028] FIG. **7** is a cross-sectional view along line **7-7** of FIG. **6**.
[0029] FIG. **8** is a partially broken away, isometric view of the detector support of FIG. **2** including a detector positioned therein, according to an exemplary embodiment of the disclosure.
[0030] FIG. **9** is a partially broken away, isometric view of the detector support of FIG. **2** including a detector positioned therein and in a bagged configuration, according to an exemplary embodiment of the disclosure.
[0031] FIG. **10** is a partially broken away, isometric view of the detector support of FIG. **2** including a detector positioned therein and being disengaged from a grid handle, according to an exemplary embodiment of the disclosure.
DETAILED DESCRIPTION OF THE DRAWINGS
[0032] One or more specific embodiments will be described below. In an effort to provide a concise description of these embodiments, all features of an actual implementation may not be described in the specification. It should be appreciated that in the development of any such actual implementation, as in any engineering or design project, numerous implementation-specific decisions must be made to achieve the developers specific goals, such as compliance with system-related and business-related constraints, which may vary from one implementation to another. Moreover, it should be appreciated that such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure.
[0033] When introducing elements of various embodiments of the present invention, the articles “a,” “an,” “the,” and “said” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Furthermore, any numerical examples in the following discussion are intended to be non-limiting, and thus additional numerical values, ranges, and percentages are within the scope of the disclosed embodiments.
[0034] In FIG. **1**, an exemplary embodiment is illustrated of a mobile x-ray device **10** constructed according to the disclosure. The mobile x-ray device **10** includes a chassis **12** that defines a body **13** including a number of wheels **14** attached to a lower surface **16** of the body **13** in order to allow the mobile x-ray device **10** to be moved over a surface by a technologist. The wheels **14** can include a pair of large support wheels **18** located on an axle (not shown) directly secured to the chassis **12**, and one or more directional wheels **20** affixed to the chassis **12** in a pivotal manner, such as casters **22**, to facilitate the movement of the chassis **12** in a desired direction. The chassis **12** can additionally include a drive handle **24** extending outwardly from the chassis **12** and graspable by a technician in order to manually direct the movement of the chassis **12** where desired. In an alternative embodiment or as an addition to the above embodiment, the chassis **12** may include a suitable motor (not shown) that is operably connected to the wheels **14**, such as to the support wheels **18**, in order to provide motorized movement capability to the chassis **12**.
[0035] The body **13** of the chassis **12** defines a rear end **28** and a front end **30**. Within the rear end **28** the body **13** encloses a number of operative computers, processing units and/or systems (not shown) as are well-known for use in the mobile x-ray device **10** such as the processing units and/or operative systems for the obtaining and processing of x-ray data to provide x-ray images using the mobile x-ray device **10**. The front end **30** defines a platform **32** that supports that supports a vertical column **34** that can be moved as desired to position an x-ray emitter **52** attached to the column **34** where necessary to obtain the x-ray images.
[0036] The column **34** can optionally be telescopic and includes a lower fixed portion **36** disposed on the platform **32**, where the fixed portion **36** is attached to the platform **32** by a suitable rotation

mechanism (not shown) by a rotational collar **37** secured to the fixed portion **36** and rotatable with respect to the platform **32**, that enables the fixed portion **36** to rotate with regard to the platform **32** along a vertical axis extending upwardly from the platform **32** through the fixed portion **36**.

[0037] The column **34** additionally includes an upper telescopic portion **38** that is moveably attached to the fixed portion **36**, such by a number of bearings (not shown) attached to the telescopic portion **38** and moveably disposed within guides (not shown) formed in the fixed portion **36**. The upper telescopic portion **38** can move vertically with respect to the fixed portion **36**.

[0038] Opposite the lower fixed portion **36**, the upper telescopic portion **38** supports a horizontal telescoping arm **40**. The telescoping arm **40** is movable vertically with regard to the telescopic portion **38**. In an exemplary embodiment, the arm **40** is engaged with the telescopic portion **38** by a number of bearings (not shown) attached to the telescoping arm **40** and moveably disposed within guides (not shown) formed in the telescopic portion **38**.

[0039] The telescoping arm **40** includes a fixed section **42** secured to a carriage **44** that is movably disposed on the telescopic portion **38**, such as by bearings (not shown) formed on the carriage **44** and engaged in guides (not shown) in the telescopic portion **38**, to provide the vertical movement of the telescoping arm **40** relative to the telescopic portion **38** of the column **34**. A number of independently moveable sections **46**,**48** are secured to the fixed section **44** and can be selectively moved in a horizontal direction with regard to the fixed section **44** and one another to extend and retract the telescoping arm **40** relative to the telescopic portion **38**.

[0040] Opposite the telescoping portion **38** the telescoping arm **40** supports a head assembly **50** on the outermost moveable section **48**. In the illustrated exemplary embodiment of FIG. **1**, the head assembly **50** for obtaining x-ray images. The head assembly **50** in the illustrated embodiment includes an x-ray source or emitter **52** and a collimator **54**, and is secured to the moveable section **48** in any suitable manner that enables the head assembly **50** to pivot and/or rotate relative to the moveable section **48** in order to position the emitter **52** where necessary to obtain the desired x-ray images.

[0041] The various components of the telescopic column **32**, i.e., the fixed portion **36** and the telescopic portion **38**, and the telescoping arm **40**, i.e., the fixed section **44** and the moveable sections **46**,**48**, are each formed of a material that is sufficiently rigid to support the various components of the x-ray device **10** that are attached thereto, while also be able to be formed with a hollow interior to enable various wiring and other operational connections between body **13** and the emitter **52**, such as through an aperture (not shown) in the platform **32** beneath the column **34**, for the operation of the x-ray device **10** to be made completely within the interior of the x-ray device **10**.

[0042] The mobile x-ray device **10** additionally includes a user console/display **56** on the body **13** in front of the drive handle **24** for the display of x-ray images obtained by the x-ray device **10** and/or for the operational control of the emitter **52** to obtain the x-ray images. In other embodiments, the console **56** can be disposed on an arm (not shown) that can be grasped by the user/technician to control the movement of the x-ray device **10**. The user console **56** can be rotated with regard to the arm between 0° and 90° on the horizontal axis between use and storage positions, and additionally can rotate between −180° and +180° on the vertical axis.

[0043] The rear end **28** additionally includes a storage bin **60** extending outward from the rear end **28** below the drive handle **24**. The storage bin **60** is configured to retain a number of detectors **62** therein, each of the detectors **62** operably connected, e.g. via a wired connection (not shown) or wirelessly, to the operative systems disposed within the body **13** in order to able to be utilized with the x-ray device **10** in an imaging procedure. The bin **60** includes spaces therein for detectors **62** of differing sizes in order to hold the detectors **62** in a manner that prevents damage to the detectors **62** from outside contact and contact with one another, while allowing the detectors **62** to be connected to a power source within the chassis **12** in the bin **60** and allowing easy access to the detectors **62** disposed within the bin **60**.

[0044] Looking now at the illustrated exemplary embodiment of FIGS. **2-8**, the drive handle **24** includes a pair of arms **66** that extend outwardly from the chassis **12** on each side of the body **13**. Opposite the body **13**, the arms **66** include a bar **68** extending therebetween. The angle of the arms **66** in relation to the body **13** positions the bar **68** at an ergonomically acceptable location where an individual can easily grasp the bar **68** to readily control the movement of the mobile x-ray device **10** while also providing sufficient room for the individual to walk normally behind the mobile x-ray device **10** without inadvertently contacting, e.g., kicking, the mobile x-ray device **10**.
[0045] In order to enable the mobile x-ray device **10** to facilitate the placement of a detector **62**, e.g., selected from within the bin **60**, into a protective bag **64** (FIG. **9**) for use in an imaging procedure, the mobile x-ray device **10** includes a detector support **100**. The detector support **100** is positioned between the arms **66** of the drive handle **24** and below the bar **68** to allow for easy access to and use of the support **100**. As best shown in FIGS. **2-5**, the detector support **100** includes a sleeve **70** having a pair of securing plates **72** extending outwardly from positions spaced inwardly from opposite ends **74**,**76** of rear side wall **78** of the sleeve **70**. The securing plates **72** can extend outwardly from the sleeve **70** at any desired angle, but in the illustrated exemplary embodiment the securing plates **72** extend outwardly at a downward angle relative to the sleeve **70**. This downward angle enables the securing plates **72** to be affixed to the adjacent arms **66** in a horizontal position, thereby facilitating the engagement of the securing plates **72** with the arms **66**, while positioning the sleeve **70** at an angle corresponding to the angle of the arms **66** to prevent the arms **66** from interfering with the insertion of a detector **62** with or without a grid handle **104** (FIG. **10**) thereon into the sleeve **70**. Additionally, the location of the securing plates **72** spaced inwardly from the ends **74**,**76** enables the securing plates **72** to be attached to the arms **66** in a configuration that aligns the ends of the sleeve **70** with the outer ends or sides of the arms **66**, such that the sleeve **70** does not extend beyond the profile of the drive handle **24**. The securing plate **72** can be affixed to the arms **66** in any suitable manner, and in the illustrated exemplary embodiment of FIG. **4** the securing plates **72** include a number of channels **80** formed therein. The channels **80** are aligned with bores (not shown) disposed in the arms **66** and can receive fasteners (not shown) therein to secure the securing plates **72** to the arms **66**.
[0046] Between the securing plates **72** the detector support **100** can optionally include a support surface **82**. The support surface **82** extends between the securing plates **72** and the sleeve **70** to form a shelf located between the arms **66** and forward of the sleeve **70** on which a number of items **84** to be employed by the user/technician during the operation of the mobile x-ray device **10**. To prevent items positioned thereon from rolling or otherwise falling off of the support surface **82**, the support surface **82** is bounded by the side wall **78** of the sleeve **70**, the securing plate **72** and a lip **86** located along the support surface **82** opposite the sleeve **70** and extending between the securing plates **72**. Further, the support surface **82** is oriented between the securing plates **72** such that the support surface **82** is disposed in a generally flat, horizontal position when the securing plates **72** are secured to the arms **66**. Also, in another exemplary embodiment, the location of the attachment of the securing pates **72** to the arms **66** disposes the support surface **82** at a location relative to the bar **68** where the support surface **82** and any items thereon can be readily accessed either under or over the bar **68**.
[0047] Looking now at the illustrated exemplary embodiment of FIGS. **2-5** and **8-9**, the sleeve **70** is disposed along a rear edge of the arms **66**, so as not to obscure or interfere with the user console/display **56**, and within the boundary or width defined by the outer side edges of the arms **66**, to maintain the sleeve **70** within the frame defined by the chassis **12**/body **13**. The sleeve **70** is formed with a bottom wall **88**, the rear side wall **78** that extends upwardly from the bottom wall **88**, a front side wall **90** extending upwardly from the bottom wall **88** opposite the rear side wall **78**, and a pair of end walls **92**,**94** extending upwardly from the bottom wall **88** and joining the adjacent edges of the rear side wall **78** and the front side wall **90**. The sleeve **70** defines an interior **96** accessible through an open top end **98** that can receive the entirety of one side edge **103** of a

detector **62** therein. The space defined by the interior **96** can be selected as desired and is capable of enabling the entire thickness or width and length of one side edge **103** of the detector **62**, again with or without the grid handle **104** (FIG. **10**), to be placed therein. In the illustrated exemplary embodiment, the bottom wall **88** is formed to be flat, or parallel in orientation to the support surface **82**. Further, the open top end **98** is oriented at an angle with regard to the bottom wall **88** and support surface **82**, such as to be aligned with the orientation of the arms **66**. In the exemplary embodiment best shown in FIGS. **4** and **5**, the height of the front side wall **90** from the bottom wall **88** is shorter than the height of the rear side wall **78**, with a corresponding slope formed on each of the end walls **92**,**94** between the rear side wall **78** and the front side wall **90** at the open top end **96**. In addition, to facilitate the cleaning of the interior **96**, the bottom wall **88** can be formed with one or more apertures **97** therein to allow smaller particulate matter and liquids to pass through the bottom wall **88**. The support surface **82** can also include apertures **99** therein for this purpose. In an alternative embodiment, the support surface **82** and/or the bottom wall **88** can be formed with a slope (not shown) to direct any small particulate matter and/or liquids towards the apertures **97**,**99** disposed at the lowest point of the slope.

[0048] As best shown in FIGS. **4-7**, the bottom wall **88** and the open top end **98** are spaced from one another a sufficient length that the entire length of each of the rear side wall **78** and front side wall **90** can engage the opposed surfaces **101**,**102** of the detector **62** with or without a grid handle **104** along their entire length in positions where an entire edge **103** of the detector **62** with or without a grid handle **104** is securely retained within the interior **96** of the sleeve **70**. Further, the spacing of the rear side wall **78** and the front side wall **90** by the end walls **92**,**94** is greater than the thickness of the detector **62**, with or without a grid handle **104**, such that in the illustrated exemplary embodiment the detector **62** can be pivoted and/or angled within the interior **96** along the edge **103** to an angle away from the rear side wall **78**. In this position, the rear side wall **78** and the front side wall **90** can securely engage opposed surfaces **101**,**102** of the detector **62** with or without a grid handle **104** while positioning enough of the detector **62** with or without a grid handle **104** outside of the sleeve **70** to allow the user/technician to readily position a protective bag **64** around the exposed portion of the detector **62** with or without a grid handle **104** in a hands-free manner, i.e., without needing to manually hold any portion of the detector **62** with or without a grid handle **104**. When the open end of the bag **64** has been positioned around the exposed portion of the detector **62** with or without a grid handle **104**, as shown in FIG. **9**, the detector **62** with or without a grid handle **104** can be removed from the sleeve **70** and inverted in order to insert the closed end of the bag **64** and the detector **62** with or without a grid handle **104** disposed therein into the sleeve **70**. In this position, the user/technician can close the open end of the bag **64** around the detector **62** with or without a grid handle **104** in a hands-free manner with the detector **62** with or without a grid handle **104** and bag **64** supported within the sleeve **70** to prepare the detector **62** for use in an imaging procedure. After use in an imaging procedure, this process can be reversed, i.e., placing the detector **62** with or without a grid handle **104** and closed bag **64** in the sleeve **70** to open the bag **64**, removing the detector **62** with or without a grid handle **104** and open bag **64** from the sleeve **70**, inverting the detector **62** with or without a grid handle **104** and open bag **64**, and reinserting the detector **62** with or without a grid handle **104** and open bag **64** into the sleeve **70** to allow for hands-free removal of the bag **64** from the detector **62** with or without a grid handle **104**.

[0049] Further, the areas of the surfaces **101**,**102** (FIG. **8**) of the detector **62** that are exposed when the detector **62** with or without a grid handle **104** is positioned within the interior **96** of the sleeve **70** allows for easy cleaning and drying of the surfaces **101**,**102**, particularly after use of the detector **62** with or without a grid handle **104** in an imaging procedure and removal of the bag **64**, again in hands-free manner, i.e., without having to manually hold any portion of the detector **62** with or without a grid handle **104** during the cleaning and drying process. Also, with regard to FIG. **10**, the width of the interior **96** being greater than that of the detector **62** allows for the assembly or disassembly of a grid handle **104** from the detector **62** while the detector **62** and grid handle **104** are securely held within the interior **96** of the detector support **100**. In one particular exemplary embodiment of the detector support **100** to facilitate this use of the detector support **100**, the front side wall **90** has a height from the bottom wall **88** of from about 2″ to about 5″, and more specifically from about 3″ to about 4″, and most particularly about 3.5″.

[0050] Referring now to FIGS. **2-3** and **6-9**, the front side wall **90** of the sleeve **70** can include a cord management system **200**. The cord management system **200** includes a cord wrap **202** and a switch holder **204**. In the illustrated exemplary embodiment, the cord wrap **202** is disposed adjacent side wall **92** and the switch holder **204** is disposed adjacent side wall **94**, though other configurations for the cord wrap **202** and/or the switch holder **204** are within the scope of the present disclosure.

[0051] The cord wrap **202** is secured to the front side wall **90** opposite the interior **96**, such as by an adhesive or mechanical fastener (not shown), e.g., one or more screws, engaged between the cord wrap **202** and the front side wall **90**. The cord wrap **202** includes a narrow end **206** disposed adjacent the front side wall **90** and a wide end **208** extending outwardly from the narrow end **206** opposite the front side wall **90**. The wide end **208** has an outer periphery dimension that extends at least partially beyond the narrow end **206** to provide an attachment location for the cord **210** connecting the X-ray exposure switch **212** to the body **13**. In the illustrated exemplary embodiment, the wide end **208** has an oval shape with a perimeter **214** that defines a channel **216** between the wide end **208** and the front side wall **90**. The cord **210** can be disposed around the narrow end **206** between the wide end **208** and the front side wall **90** to maintain the cord **210** in a storage position adjacent the front side wall **90**, as shown in FIGS. **2** and **9**.

[0052] The switch holder **204** is secured to the front side wall **90** opposite the interior **96**, such as by an adhesive or mechanical fastener (not shown), e.g., one or more screws, engaged between the switch holder **204** and the front side wall **90**. In the illustrated exemplary embodiment, the switch holder **204** includes a base **218** adjacent the front side wall **90** and a pair of retaining arms **220** disposed on and extending outwardly away from opposed sides of the base **218** to define a retaining space **222** therebetween. The retaining arms **220** extend from the base **218** inwardly towards one another and are separated opposite the base **218** by a slot **224** disposed between the retaining arms **220**. The retaining space **222** can also optionally be defined by the retaining arms **220** including a taper or narrowing from a wide upper end **226** to a narrow lower end **228**. The dimensions of the retaining space **222** defined by the retaining arms **220** enables the activation switch **212** to be placed into the retaining space **222** through the upper end **226** to engage or rest upon the lower end **228**, The cord **210** connecting the activation switch **212** to the body **13** can be positioned within the slot **224** to allow the activation switch **212** to be fully positioned within the switch holder **204** without interference of the cord **210**.

[0053] When using the switch holder **204**, to prevent the cord **210** and/or switch **212** from inadvertently being positioned within the interior **96** of the sleeve **70** the switch holder **204** includes a guide **230**. The guide **230** extends upwardly from the base **218** above the front side wall **90**. Thus, the guide **230** functions to stop the switch **212** from being positioned above the interior **96** of the sleeve **70** when moved towards the switch holder **204** and directs the cord **210** and/or the activation switch **212** into the switch holder **204**. Additionally, in an alternative embodiment multiple guide **230** can be disposed on and spaced along the length of the front side wall **90** adjacent the open upper end **92** to provide a similar function to direct the detector **60** into the interior **96** of the sleeve **70**.

[0054] The detector support **100** including the sleeve **70**, and optionally including the support surface **82** and one or both of the cord wrap **202** and switch holder **204** of the cord management system **200**, can be formed from any suitable material, such as a suitable metal or plastic material. The detector support **100** disclosed herein can be formed as a unitary member of from different component members secured to one another. Further, the detector support **100** including the sleeve **70** can be utilized as a retrofit component to an existing mobile x-ray device **10**, where the sleeve

70 is secured to a drive handle **24** of the mobile x-ray device **10** adjacent a shelf (not shown) already disposed on the drive handle **24**, or the shelf directly.

[0055] It is understood that the aforementioned compositions, apparatuses and methods of this disclosure are not limited to the particular embodiments and methodology, as these may vary. It is also understood that the terminology used herein is for the purpose of describing particular exemplary embodiments only, and is not intended to limit the scope of the present disclosure which will be limited only by the appended claims.

Claims

1. A detector support for a mobile x-ray device, the detector support comprising: a. a sleeve defining an interior adapted to receive an edge of a detector therein; and b. a pair of securing plates extending outwardly from the sleeve and adapted to be secured to the mobile x-ray device.

2. The detector support of claim 1, wherein the sleeve comprises: a. a bottom wall; b. a rear side wall extending upwardly from the bottom wall; c. a front side wall extending upwardly from the bottom wall opposite the rear side wall; d. a pair of end walls extending upwardly from the bottom wall and joining opposed ends of the rear side wall and the front side wall, and e. an open upper end disposed opposite the bottom wall.

3. The detector support of claim 2, wherein the rear side wall and front side wall are disposed at an angle with regard to the bottom wall.

4. The detector support of claim 3, wherein the front side wall is shorter than the rear side wall.

5. The detector support of claim 4, wherein the front side wall has a height between about 2″ and about 5″.

6. The detector support of claim 1, wherein the securing plates extend at a downward angle relative to the rear side wall.

7. The detector support of claim 6, further comprising a support surface disposed between the sleeve and the pair of securing plates.

8. The detector support of claim 1, further comprising a cord management system disposed on the front side wall opposite the interior.

9. The detector support of claim 8, wherein the cord management system comprises a. a cord wrap disposed on the front side wall; and b. a switch holder disposed on the front side wall.

10. A mobile x-ray device comprising: a. a body including a number of wheels rotatably attached thereto and housing operative systems for the obtaining and processing of x-ray data to provide x-ray images; b. a moveable vertical column and horizontal arm operably connected to the body and including an x-ray emitter opposite the body; c. a detector configured to receive x-rays from the x-ray emitter and operably connected to the operative systems to transmit x-ray data from the detector to the operative systems; d. a drive handle disposed on the body; and e. a detector support comprising: i. a sleeve defining an interior adapted to receive an edge of a detector therein; and ii. a pair of securing plates extending outwardly from the sleeve and secured to the drive handle.

11. The mobile x-ray device of claim 10, wherein the sleeve comprises: a. a bottom wall; b. a rear side wall extending upwardly from the bottom wall; c. a front side wall extending upwardly from the bottom wall opposite the rear side wall; d. a pair of end walls extending upwardly from the bottom wall and joining opposed ends of the rear side wall and the front side wall, and e. an open upper end disposed opposite the bottom wall.

12. The mobile x-ray device of claim 11, wherein the front side wall is shorter than the rear side wall.

13. The mobile x-ray device of claim 12, wherein the front side wall has a height between about 2″ and about 5″.

14. The mobile x-ray device of claim 10, further comprising a support surface disposed between the sleeve and the pair of securing plates.

15. The mobile x-ray device of claim 10, further comprising a cord management system disposed on the front side wall opposite the interior.
16. The mobile x-ray device of claim 15, wherein the cord management system comprises a. a cord wrap disposed on the front side wall; and b. a switch holder disposed on the front side wall.
17. The mobile x-ray device of claim 10, wherein the sleeve does not extend beyond outer sides of the drive handle.
18. A method of supporting a detector on a mobile x-ray device, the method comprising the steps of: a. providing a mobile x-ray device comprising: i. a body including a number of wheels rotatably attached thereto and housing operative systems for the obtaining and processing of x-ray data to provide x-ray images; ii. a moveable vertical column and horizontal arm operably connected to the body and including an x-ray emitter opposite the body; iii. a detector configured to receive x-rays from the x-ray emitter and operably connected to the operative systems to transmit x-ray data from the detector to the operative systems; iv. a drive handle disposed on the body; and v. a detector support comprising: 1. a sleeve defining an interior adapted to receive an edge of a detector therein; and 2. a pair of securing plates extending outwardly from the sleeve and secured to the drive handle; and b. placing an edge of the detector within the sleeve to support the detector within the sleeve in a hands-free manner; and c. placing a protective bag around the detector.
19. The method of claim 18, further comprising the steps of: a. performing an imaging procedure with the detector within the protective bag; b. placing the detector and protective bag within the sleeve; and c. removing the protective bag from the detector.
20. The method of claim 18, further comprising cleaning the detector within the sleeve after removing the protective bag.